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Project Report

Project Title

Daily Habit Tracker with Gamification

1. Project Overview

Purpose & Concept

This project is designed to help users build and maintain positive habits by tracking their daily routines and rewarding them for consistency. By integrating gamified elements, such as streak tracking, badge collection, and level progression, it encourages sustained user engagement and motivation.

Tech Stack

- **Frontend:** React — Lightweight, reactive UI for interactive habit tracking and visual analytics.
 - **Backend:** Spring Boot — Robust and scalable API providing business logic and user data management.
 - **Database:** MongoDB — Flexible document-oriented storage ideal for user habits, streaks, sessions, and badge metadata.
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2. Core Modules

a) Habit Calendar

- Visual calendar interface displaying daily habit completion.
- Easy marking and editing of habits.
- Supports overview of current streaks and historical performance.

b) Badges & Level System

- Automatic awarding of badges based on user milestones (e.g., “7-day streak”, “Level 5”).
- Level-based progression unlocks new features or visuals—adds a sense of achievement.
- Badge gallery for users to view earned rewards.

c) Progress Statistics

- **Visual Analytics:** Charts (bar, line, etc.) showing daily, weekly, and monthly habit performance.
 - **Weekly Breakdown:** Comparative overview (e.g., “Habit A: 5/7 days completed this week”).
 - **Streak Insights:** Longest streaks, current streak, potential streak-breaking days, and success probabilities.
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3. Evaluation Metrics

- **Streak Tracking:** Detect and record consecutive days of consistency.
 - **Level Progression:** Define XP thresholds per level; award XP for habit completions.
 - **Badge Milestones:** Badge triggers for reaching new levels, streak milestones (e.g., 3-day, 7-day, 30-day).
 - **Analytics Dashboard:**
 - Charts for tracking habit adherence over time.
 - Weekly summaries for quick performance snapshots.
 - Highlighting patterns (like best days, lulls) to encourage behavior adjustments.
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4. User Journey Example

1. **Onboarding:**
 - User signs up, lists habits they want to track (e.g., “Read”, “Workout”, “Meditate”).
 2. **Daily Use:**
 - Each day, the user checks off completed habits in the Habit Calendar.
 3. **Progress Visualized:**
 - Calendar shows marked days; dashboard updates with new visuals.
 4. **Reward Prompted:**
 - After completing a streak or hitting XP target, user receives a badge or levels up.
 5. **Engagement Reinforced:**
 - User visits dashboard to see their stats, reflect on patterns, and strive for improvement.
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5. Technical Architecture

1. **Frontend (React):**
 - **Components:** Calendar, Badge Gallery, Analytics Dashboard, Habit List.
 - **State Management:** Redux or Context API for handling habit data and user session.
 - **API Interaction:** Axios or Fetch to communicate with the Spring Boot backend.
2. **Backend (Spring Boot):**
 - REST endpoints for CRUD operations on habits, stats, badges, and user profiles.
 - Business logic for computing streaks, XP, levels, and badge eligibility.
 - Integration with MongoDB for persistent user and habit data.
3. **Database (MongoDB):**
 - Collections for Users, Habits, Completions, Badges, Levels, and Analytics Data.
 - Flexible schema to store habit completion logs, badge metadata, and XP thresholds.

6. Benefits & Use Cases

- **Motivational Tool:** Reinforces daily habit formation with visual feedback and rewards.
- **Behavioral Insight:** Analytics help users understand and refine their routines.
- **Wide Applicability:** Useful for personal wellness, education, fitness, productivity, mental health tracking.

7. Potential Enhancements (Future Scope)

- **Social Sharing / Leaderboards:** Let users compare streaks or achievements with friends.
- **Reminders & Notifications:** Push or in-app alerts to prompt habit completion.
- **Custom XP & Badge Creation:** Users design personalized challenges and rewards.
- **Gamified Levels & Unlocks:** Offer themes, avatars, or bonus content as users advance.
- **Cross-Platform Support:** Mobile-responsive UI or dedicated mobile app.